

Question:

A 61-year-old woman is brought to an urgent care clinic by her husband with an altered mental status. The patient’s husband says that her symptoms onset acutely 24 hours ago. He says she suddenly started to feel excessively drowsy and quickly became altered. Past medical history is significant for renal transplant 18 months ago, well-managed with immunosuppressive medication. The vital signs include: temperature 39.4°C (103.0°F), blood pressure 85/50 mm Hg, pulse 135/min and respirations 24/min. On physical examination, the patient is arousable but non-responsive to commands. Oriented x 0. Glasgow Coma Scale (GCS) score is 10/15. Multiple black skin lesions are present on the trunk. Lesions are approximately 4 cm is diameter and round with a necrotic center forming an eschar. Laboratory findings are significant for the following:Hemoglobin 14.2 g/dLWBC count 3,700/μLNeutrophils 22%Lymphocytes 52%Monocytes 17%Eosinophils 5%Basophils 4%Platelets 179,000/μLBUN 15 mg/dLCreatinine 0.8 mg/dLBlood cultures are positive for Pseudomonas aeruginosa. Which of the following factors is most likely responsible for this patient’s condition?Answer Choices:A. Failure of immunosuppressive therapyB. Decreased interleukin-2 levelsC. DiGeorge SyndromeD. Decreased phagocytic cell count

Reasoning + Answer:

<think>To answer this question, we need to consider the patient's symptoms and risk factors. The patient is a 61-year-old woman with a past medical history of renal transplant, which makes her more susceptible to infections, particularly those caused by pseudomonas aeruginosa. The patient's symptoms, including altered mental status, fever, and black skin lesions, are consistent with a possible infection, such as necrotizing fasciitis, which is a known complication of pseudomonas aeruginosa infections. The patient's laboratory findings, including a low white blood cell count and positive blood cultures for pseudomonas aeruginosa, also support this diagnosis. The most likely cause of the patient's condition is a failure of immunosuppressive therapy, which has left her vulnerable to infection. </think><answer>Failure of immunosuppressive therapy</answer>

